

SETTING UP MULTI-AREA OSPF

Period 6–8

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CCNP Lab 2



Purpose

The purpose of this lab is to practice setting up and managing a network using multi-area OSPF with both IPv4 and IPv6. By configuring routers and Layer 3 switches, the lab shows how OSPF can be used to organize larger networks into areas, which makes routing more efficient. It also helps with understanding how routing protocols adapt in different environments. Additionally, we can see how multi-area OSPF reduces routing table size and leads to more manageable and resilient routing in complex networks.

Background Information

The protocol Open Shortest Path First (OSPF) is used on networks to find the most effective routes to destinations by dynamically exchanging routing information. Since it is an open standard protocol, a wide variety of routers can use it. Additionally, OSPF uses the Shortest Path First (SPF) algorithm, a network routing technique that determines the quickest route between a point (such as a router) and every other device in a topology. When the topology shifts, the new “shortest path” between devices is calculated. The biggest benefit of OSPF is easier scalability for a network. With OSPF, networks can be divided into different areas, which reduces the impact of changes within one area on the rest of the network.

OSPF can be broken down into five steps:

1. Starting the process by enabling OSPF and setting router IDs. Each router needs an ID number so it can identify itself when communicating with other routers.
2. Establishing neighbor adjacencies by sending Hello messages. To decide whether neighbors are present on any lines that are OSPF-enabled, the router begins sending Hello messages on each link (interface). The router attempts to create a neighbor adjacency with any other OSPF-enabled devices on the link. Routers then look for other routers nearby that are also using OSPF. They do this by sending “Hello” messages, which are like introductions. If two routers both respond, they become neighbors and start working together.
3. A router begins sharing Link-state Advertisements, which provide the IP address of the directly linked link, once it has neighbors. Until every device in the vicinity has access to all the information, each router bombards its neighbors with Link-state Advertisements. (The Link-state Database, which contains all of the network's topology information, is where the Link-state Advertisements are stored.) The Link-state Database of every router in the same region contains identical data.
4. The Link-state Database is an input for the SPF algorithm, which calculates the best paths based on the cost of each link (cost is a path metric that is calculated using reference bandwidth/interface bandwidth)

5. Last, the router adds the best paths from the SPF tree into the routing table and then makes forwarding decisions based on the information in the routing table.

Lab Summary

In this lab, we configured a network consisting of five routers and two Layer 3 switches to run multi-area OSPF. We assigned both IPv4 and IPv6 addresses across the devices, so that each interface was properly addressed and reachable. I set up OSPF to divide the network into areas 0, 1, and 51, verified that routes were exchanged correctly between areas, and confirmed connectivity throughout the network. This allowed me to demonstrate how multi-area OSPF improves efficiency and scalability in larger networks and sent packets between all the devices.

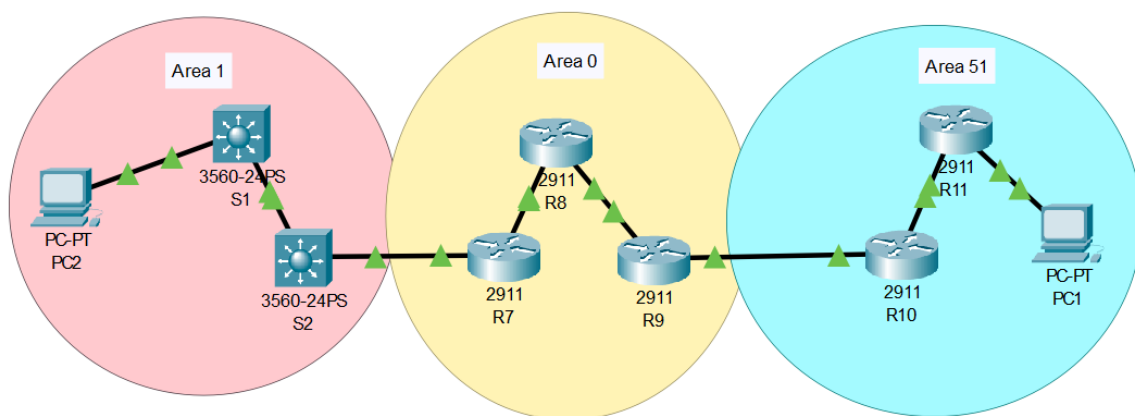
Lab Commands

`router ospf x`: This command starts the OSPF routing process on the router. x is a number that represents the local process ID (not the OSPF area number), which just identifies the OSPF process on that router.

`router-id x.x.x.x`: This sets the OSPF router ID for a device it must be unique and written in IPv4 dotted-decimal format. The router ID is how the router is identified in the OSPF domain (not an actual IP address). If you don't manually set it, OSPF picks the highest IP address on an active interface.

`ip ospf x area y`: This tells OSPF to advertise the IP network configured on an interface into the OSPF process x and associate it with area y. It enables OSPF on that interface and makes the network reachable in the OSPF domain.

Network Diagram with IP's



We used ipv6 addresses that corresponded to the ipv4 address.

For example: for router 7, we used 2001:db8:acad:3::2/64 since the ipv4 address ends in .3.2

Router 7	G/0/0 (S2 g0/2)	G0/0/1(R8 g0/0)
	192.168.3.2	192.168.4.1
Router 8	G/0/0 (R7 g0/1)	G0/0/1 (R9 g0/0)
	192.168.4.2	192.168.5.1
Router 9	G/0/0 (R8 g0/1)	G0/0/1 (R10 g0/0)
	192.168.5.2	192.168.6.1
Router 10	G/0/0 (R9 g0/1)	G0/0/1 (R11 g0/0)
	192.168.6.2	192.168.7.1
Router 11	G/0/0 (R10 g0/1)	G0/0/1 (PC)
	192.168.7.2	192.168.8.1
Switch 1	G/0/2 (PC)	G0/0/1 (S2 g0/1)
	192.168.1.1	192.168.2.1
Switch 2	G1/0/2 (R7 g0/0)	G0/0/1 (S1 g0/1)
	192.168.3.1 (area 0)	192.168.2.2 (area 51)

Area 0

Area 1

Area 51

255.255.255.252 (subnet mask)

Ports connected to PCs: 255.255.255.0 (subnet mask for ports connected to PCs)

Configurations

PC 1 to PC 2 traceroutes:

IPv4: >tracert 192.168.1.2 Tracing route to DESKTOP-TJIC15F [192.168.1.2] over a maximum of 30 hops: 1 <1 ms <1 ms <1 ms 192.168.8.1 2 <1 ms <1 ms <1 ms 192.168.7.1 3 <1 ms <1 ms <1 ms 192.168.6.1 4 <1 ms <1 ms <1 ms 192.168.5.1 5 <1 ms <1 ms <1 ms 192.168.4.1 6 1 ms 1 ms <1 ms 192.168.3.1 7 1 ms 1 ms <1 ms 192.168.2.1 8 1 ms 1 ms 1 ms DESKTOP-TJIC15F [192.168.1.2]	IPv6: >tracert 2001:db8:acad:1::2 Tracing route to 2001:db8:acad:1::2 over a maximum of 30 hops 1 <1 ms <1 ms <1 ms 2001:db8:acad:8::1 2 1 ms <1 ms <1 ms 2001:db8:acad:7::1 3 <1 ms <1 ms <1 ms 2001:db8:acad:6::1 4 1 ms 1 ms <1 ms 2001:db8:acad:5::1 5 1 ms <1 ms <1 ms 2001:db8:acad:4::1 6 1 ms 1 ms <1 ms 2001:db8:acad:3::1 7 1 ms * 1 ms 2001:db8:acad:2::1 8 9 ms 1 ms 1 ms 2001:db8:acad:1::2
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PC 2 to PC 1 ipv4 traceroute:

IPv4: >tracert 192.168.8.2 Tracing route to DESKTOP-JCHOFNQ [192.168.8.2] over a maximum of 30 hops: 1 1 ms <1 ms <1 ms 192.168.1.1 2 1 ms <1 ms <1 ms 192.168.2.2 3 <1 ms <1 ms <1 ms 192.168.3.2 4 <1 ms <1 ms <1 ms 192.168.4.2 5 <1 ms <1 ms <1 ms 192.168.5.2 6 <1 ms <1 ms <1 ms 192.168.6.2 7 <1 ms <1 ms <1 ms 192.168.7.2 8 1 ms 1 ms 1 ms DESKTOP-JCHOFNQ [192.168.8.2]	IPv6: >tracert 2001:db8:acad:8::2 Tracing route to 2001:db8:acad:8::2 over a maximum of 30 hops 1 2 ms 1 ms 2 ms 2001:db8:acad:1::1 2 1 ms <1 ms <1 ms 2001:db8:acad:2::2 3 1 ms <1 ms <1 ms 2001:db8:acad:3::2 4 <1 ms <1 ms <1 ms 2001:db8:acad:4::2 5 1 ms 1 ms <1 ms 2001:db8:acad:5::2 6 1 ms <1 ms <1 ms 2001:db8:acad:6::2 7 1 ms 1 ms <1 ms 2001:db8:acad:7::2 8 5 ms 1 ms 1 ms 2001:db8:acad:8::2
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ROUTER 7:

R7#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```
O IA 192.168.1.0/24 [110/3] via 192.168.3.1, 00:55:31, GigabitEthernet0/0/0
    192.168.2.0/30 is subnetted, 1 subnets
O IA 192.168.2.0 [110/2] via 192.168.3.1, 00:55:31, GigabitEthernet0/0/0
    192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.3.0/30 is directly connected, GigabitEthernet0/0/0
L 192.168.3.2/32 is directly connected, GigabitEthernet0/0/0
    192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.4.0/30 is directly connected, GigabitEthernet0/0/1
L 192.168.4.1/32 is directly connected, GigabitEthernet0/0/1
    192.168.5.0/30 is subnetted, 1 subnets
O 192.168.5.0 [110/2] via 192.168.4.2, 00:52:09, GigabitEthernet0/0/1
    192.168.6.0/30 is subnetted, 1 subnets
O 192.168.6.0 [110/3] via 192.168.4.2, 00:49:21, GigabitEthernet0/0/1
    192.168.7.0/30 is subnetted, 1 subnets
O IA 192.168.7.0 [110/4] via 192.168.4.2, 00:47:38, GigabitEthernet0/0/1
O IA 192.168.8.0/24 [110/5] via 192.168.4.2, 00:47:03, GigabitEthernet0/0/1
```

R7#show ipv6 route

IPv6 Routing Table - default - 11 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external
ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application

```
OI 2001:DB8:ACAD:1::/64 [110/3]
    via FE80::B28B:D0FF:FED6:C56, GigabitEthernet0/0/0
OI 2001:DB8:ACAD:2::/64 [110/2]
    via FE80::B28B:D0FF:FED6:C56, GigabitEthernet0/0/0
C 2001:DB8:ACAD:3::/64 [0/0]
    via GigabitEthernet0/0/0, directly connected
L 2001:DB8:ACAD:3::2/128 [0/0]
    via GigabitEthernet0/0/0, receive
C 2001:DB8:ACAD:4::/64 [0/0]
    via GigabitEthernet0/0/1, directly connected
L 2001:DB8:ACAD:4::1/128 [0/0]
    via GigabitEthernet0/0/1, receive
O 2001:DB8:ACAD:5::/64 [110/2]
    via FE80::B6A8:B9FF:FE47:92C0, GigabitEthernet0/0/1
O 2001:DB8:ACAD:6::/64 [110/3]
    via FE80::B6A8:B9FF:FE47:92C0, GigabitEthernet0/0/1
OI 2001:DB8:ACAD:7::/64 [110/4]
    via FE80::B6A8:B9FF:FE47:92C0, GigabitEthernet0/0/1
OI 2001:DB8:ACAD:8::/64 [110/5]
    via FE80::B6A8:B9FF:FE47:92C0, GigabitEthernet0/0/1
L FF00::/8 [0/0]
    via Null0, receive
```

R7#show ip ospf neighbor detail

Neighbor 8.8.8.8, interface address 192.168.4.2
In the area 0 via interface GigabitEthernet0/0/1
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.4.2 BDR is 192.168.4.1
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:36
Neighbor is up for 00:54:10
Index 1/2/2, retransmission queue length 0, number of retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec

Neighbor 2.2.2.2, interface address 192.168.3.1
In the area 0 via interface GigabitEthernet0/0/0
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.3.1 BDR is 192.168.3.2
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:32
Neighbor is up for 00:56:53
Index 1/1/1, retransmission queue length 0, number of retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec

R7#show ipv6 ospf neighbor detail

OSPFv3 Router with ID (7.7.7.7) (Process ID 1)

Neighbor 8.8.8.8
In the **area 0** via interface GigabitEthernet0/0/1
Neighbor: interface-id 6, link-local address FE80::B6A8:B9FF:FE47:92C0
Neighbor priority is 1, State is FULL, 6 state changes
DR is 8.8.8.8 BDR is 7.7.7.7
Options is 0x000013 in Hello (V6-Bit, E-Bit, R-Bit)
Options is 0x000013 in DBD (V6-Bit, E-Bit, R-Bit)
Dead timer due in 00:00:36
Neighbor is up for 00:37:45
Index 1/1/1, retransmission queue length 0, number of retransmission 1
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 2, maximum is 2
Last retransmission scan time is 1 msec, maximum is 1 msec

Neighbor 2.2.2.2
In the **area 0** via interface GigabitEthernet0/0/0
Neighbor: interface-id 38, link-local address FE80::B28B:D0FF:FED6:C56
Neighbor priority is 1, State is FULL, 6 state changes
DR is 7.7.7.7 BDR is 2.2.2.2
Options is 0x000013 in Hello (V6-Bit, E-Bit, R-Bit)
Options is 0x000013 in DBD (V6-Bit, E-Bit, R-Bit)
Dead timer due in 00:00:33
Neighbor is up for 00:37:51
Index 1/2/2, retransmission queue length 0, number of retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec

R7#show ip ospf interface brief

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Gi0/0/1	1	0	192.168.4.1/30	1	BDR	1/1	
Gi0/0/0	1	0	192.168.3.2/30	1	BDR	1/1	

ROUTER 8:

R8#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
Gateway of last resort is not set
O IA 192.168.1.0/24 [110/4] via 192.168.4.1, 00:47:27, GigabitEthernet0/0/0
192.168.2.0/30 is subnetted, 1 subnets
O IA 192.168.2.0 [110/3] via 192.168.4.1, 00:47:27, GigabitEthernet0/0/0
192.168.3.0/30 is subnetted, 1 subnets
O 192.168.3.0 [110/2] via 192.168.4.1, 00:47:27, GigabitEthernet0/0/0
192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.4.0/30 is directly connected, GigabitEthernet0/0/0
L 192.168.4.2/32 is directly connected, GigabitEthernet0/0/0
192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.5.0/30 is directly connected, GigabitEthernet0/0/1
L 192.168.5.1/32 is directly connected, GigabitEthernet0/0/1
192.168.6.0/30 is subnetted, 1 subnets
O 192.168.6.0 [110/2] via 192.168.5.2, 00:44:40, GigabitEthernet0/0/1
192.168.7.0/30 is subnetted, 1 subnets
O IA 192.168.7.0 [110/3] via 192.168.5.2, 00:42:56, GigabitEthernet0/0/1
O IA 192.168.8.0/24 [110/4] via 192.168.5.2, 00:42:21, GigabitEthernet0/0/1

R8#show ip ospf neighbor detail

Neighbor 9.9.9.9, interface address 192.168.5.2
In the area 0 via interface GigabitEthernet0/0/1
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.5.1 BDR is 192.168.5.2
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:33
Neighbor is up for 00:47:14
Index 1/2/2, retransmission queue length 0, number of retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec
Neighbor 7.7.7.7, interface address 192.168.4.1
In the area 0 via interface GigabitEthernet0/0/0
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.4.2 BDR is 192.168.4.1
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:32
Neighbor is up for 00:48:49
Index 1/1/1, retransmission queue length 0, number of retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec

R8#show ipv6 route

IPv6 Routing Table - default - 11 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
B - BGP, R - RIP, H - NHRP, I1 - ISIS L1
I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP
EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination
NDR - Redirect, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1
OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
a - Application
OI 2001:DB8:ACAD:1::/64 [110/4]
via FE80::B6A8:B9FF:FE01:B511, GigabitEthernet0/0/0
OI 2001:DB8:ACAD:2::/64 [110/3]
via FE80::B6A8:B9FF:FE01:B511, GigabitEthernet0/0/0
O 2001:DB8:ACAD:3::/64 [110/2]
via FE80::B6A8:B9FF:FE01:B511, GigabitEthernet0/0/0
C 2001:DB8:ACAD:4::/64 [0/0]
via GigabitEthernet0/0/0, directly connected
L 2001:DB8:ACAD:4::2/128 [0/0]
via GigabitEthernet0/0/0, receive
C 2001:DB8:ACAD:5::/64 [0/0]
via GigabitEthernet0/0/1, directly connected
L 2001:DB8:ACAD:5::1/128 [0/0]
via GigabitEthernet0/0/1, receive
O 2001:DB8:ACAD:6::/64 [110/2]

R8#show ipv6 ospf neighbor detail

OSPFv3 Router with ID (8.8.8.8) (Process ID 1)
Neighbor 9.9.9.9
In the area 0 via interface GigabitEthernet0/0/1
Neighbor: interface-id 6, link-local address FE80::B6A8:B9FF:FE01:AE50
Neighbor priority is 1, State is FULL, 6 state changes
DR is 9.9.9.9 BDR is 8.8.8.8
Options is 0x000013 in Hello (V6-Bit, E-Bit, R-Bit)
Options is 0x000013 in DBD (V6-Bit, E-Bit, R-Bit)
Dead timer due in 00:00:32
Neighbor is up for 00:34:56
Index 1/1/1, retransmission queue length 0, number of retransmission 1
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 1, maximum is 1
Last retransmission scan time is 0 msec, maximum is 0 msec
Neighbor 7.7.7.7
In the area 0 via interface GigabitEthernet0/0/0
Neighbor: interface-id 7, link-local address FE80::B6A8:B9FF:FE01:B511
Neighbor priority is 1, State is FULL, 6 state changes
DR is 8.8.8.8 BDR is 7.7.7.7
Options is 0x000013 in Hello (V6-Bit, E-Bit, R-Bit)
Options is 0x000013 in DBD (V6-Bit, E-Bit, R-Bit)
Dead timer due in 00:00:30
Neighbor is up for 00:32:24
Index 1/2/2, retransmission queue length 0, number of retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec

R8#show ip ospf int brief

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Gi0/0/1	1	0	192.168.5.1/30	1	DR	1/1	
Gi0/0/0	1	0	192.168.4.2/30	1	DR	1/1	

ROUTER 9:

R9#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

O IA 192.168.1.0/24 [110/5] via 192.168.5.1, 00:42:32, GigabitEthernet0/0/0
192.168.2.0/30 is subnetted, 1 subnets
O IA 192.168.2.0 [110/4] via 192.168.5.1, 00:42:32, GigabitEthernet0/0/0
192.168.3.0/30 is subnetted, 1 subnets
O 192.168.3.0 [110/3] via 192.168.5.1, 00:42:32, GigabitEthernet0/0/0
192.168.4.0/30 is subnetted, 1 subnets
O 192.168.4.0 [110/2] via 192.168.5.1, 00:42:32, GigabitEthernet0/0/0
192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.5.0/30 is directly connected, GigabitEthernet0/0/0
L 192.168.5.2/32 is directly connected, GigabitEthernet0/0/0
192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.6.0/30 is directly connected, GigabitEthernet0/0/1
L 192.168.6.1/32 is directly connected, GigabitEthernet0/0/1
192.168.7.0/30 is subnetted, 1 subnets
O IA 192.168.7.0 [110/2] via 192.168.6.2, 00:39:00, GigabitEthernet0/0/1
O IA 192.168.8.0/24 [110/3] via 192.168.6.2, 00:38:25, GigabitEthernet0/0/1

R9#show ipv6 route

IPv6 Routing Table - default - 11 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
B - BGP, R - RIP, H - NHRP, I1 - ISIS L1
I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP
EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination
NDR - Redirect, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1
OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
a - Application

OI 2001:DB8:ACAD:1::/64 [110/5]
via FE80::B6A8:B9FF:FE47:92C1, GigabitEthernet0/0/0
OI 2001:DB8:ACAD:2::/64 [110/4]
via FE80::B6A8:B9FF:FE47:92C1, GigabitEthernet0/0/0
O 2001:DB8:ACAD:3::/64 [110/3]
via FE80::B6A8:B9FF:FE47:92C1, GigabitEthernet0/0/0
O 2001:DB8:ACAD:4::/64 [110/2]
via FE80::B6A8:B9FF:FE47:92C1, GigabitEthernet0/0/0
C 2001:DB8:ACAD:5::/64 [0/0]
via GigabitEthernet0/0/0, directly connected
L 2001:DB8:ACAD:5::2/128 [0/0]
via GigabitEthernet0/0/0, receive
C 2001:DB8:ACAD:6::/64 [0/0]
via GigabitEthernet0/0/1, directly connected
L 2001:DB8:ACAD:6::1/128 [0/0]
via GigabitEthernet0/0/1, receive
OI 2001:DB8:ACAD:7::/64 [110/2]
via FE80::521C:B0FF:FE2D:7100, GigabitEthernet0/0/1
OI 2001:DB8:ACAD:8::/64 [110/3]
via FE80::521C:B0FF:FE2D:7100, GigabitEthernet0/0/1
L FF00::/8 [0/0]
via Null0, receive

R9#show ip ospf neighbor detail

Neighbor 10.10.10.10, interface address 192.168.6.2
In the area 0 via interface GigabitEthernet0/0/1
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.6.1 BDR is 192.168.6.2
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:39
Neighbor is up for 00:40:09
Index 1/2/2, retransmission queue length 0, number of retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec

Neighbor 8.8.8.8, interface address 192.168.5.1
In the area 0 via interface GigabitEthernet0/0/0
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.5.1 BDR is 192.168.5.2
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:33
Neighbor is up for 00:43:02
Index 1/1/1, retransmission queue length 0, number of retransmission 1
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 1, maximum is 1
Last retransmission scan time is 0 msec, maximum is 0 msec

R9#show ipv6 ospf neighbor detail

OSPFv3 Router with ID (9.9.9.9) (Process ID 1)
Neighbor 10.10.10.10
In the area 0 via interface GigabitEthernet0/0/1
Neighbor: interface-id 6, link-local address FE80::521C:B0FF:FE2D:7100
Neighbor priority is 1, State is FULL, 6 state changes
DR is 10.10.10.10 BDR is 9.9.9.9
Options is 0x000013 in Hello (V6-Bit, E-Bit, R-Bit)
Options is 0x000013 in DBD (V6-Bit, E-Bit, R-Bit)
Dead timer due in 00:00:32
Neighbor is up for 00:33:34
Index 1/1/1, retransmission queue length 0, number of retransmission 1
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 1, maximum is 1
Last retransmission scan time is 0 msec, maximum is 0 msec

Neighbor 8.8.8.8
In the area 0 via interface GigabitEthernet0/0/0
Neighbor: interface-id 7, link-local address FE80::B6A8:B9FF:FE47:92C1
Neighbor priority is 1, State is FULL, 6 state changes
DR is 9.9.9.9 BDR is 8.8.8.8
Options is 0x000013 in Hello (V6-Bit, E-Bit, R-Bit)
Options is 0x000013 in DBD (V6-Bit, E-Bit, R-Bit)
Dead timer due in 00:00:38
Neighbor is up for 00:30:54
Index 1/2/2, retransmission queue length 0, number of retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec

R9#show ip ospf int brief

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs F/C
Gi0/0/1	1	0	192.168.6.1/30	1	DR	1/1
Gi0/0/0	1	0	192.168.5.2/30	1	BDR	1/1

ROUTER 10:

R10#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

O IA 192.168.1.0/24 [110/6] via 192.168.6.1, 00:25:57, GigabitEthernet0/0/0
192.168.2.0/30 is subnetted, 1 subnets
O IA 192.168.2.0 [110/5] via 192.168.6.1, 00:25:57, GigabitEthernet0/0/0
192.168.3.0/30 is subnetted, 1 subnets
O 192.168.3.0 [110/4] via 192.168.6.1, 00:25:57, GigabitEthernet0/0/0
192.168.4.0/30 is subnetted, 1 subnets
O 192.168.4.0 [110/3] via 192.168.6.1, 00:25:57, GigabitEthernet0/0/0
192.168.5.0/30 is subnetted, 1 subnets
O 192.168.5.0 [110/2] via 192.168.6.1, 00:25:57, GigabitEthernet0/0/0
192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks

R10#show ip ospf neighbor detail

Neighbor 9.9.9.9, interface address 192.168.6.1
In the area 0 via interface GigabitEthernet0/0/0
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.6.1 BDR is 192.168.6.2
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:30
Neighbor is up for 00:27:23
Index 1/1/1, retransmission queue length 0, number of retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec

Neighbor 11.11.11.11, interface address 192.168.7.2
In the area 1 via interface GigabitEthernet0/0/1
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.7.2 BDR is 192.168.7.1
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:36
Neighbor is up for 00:26:41
Index 1/1/2, retransmission queue length 0, number of retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec

R10#show ipv6 ospf neighbor detail

OSPFv3 Router with ID (10.10.10.10) (Process ID 1)
Neighbor 9.9.9.9
In the area 0 via interface GigabitEthernet0/0/0
Neighbor: interface-id 7, link-local address FE80::B6A8:B9FF:FE01:AE51
Neighbor priority is 1, State is FULL, 6 state changes
DR is 10.10.10.10 BDR is 9.9.9.9
Options is 0x000013 in Hello (V6-Bit, E-Bit, R-Bit)
Options is 0x000013 in DBD (V6-Bit, E-Bit, R-Bit)
Dead timer due in 00:00:38
Neighbor is up for 00:20:32
Index 1/1/1, retransmission queue length 0, number of retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec

Neighbor 11.11.11.11
In the area 1 via interface GigabitEthernet0/0/1
Neighbor: interface-id 6, link-local address FE80::6EDD:30FF:FEB6:6030
Neighbor priority is 1, State is FULL, 6 state changes
DR is 10.10.10.10 BDR is 11.11.11.11
Options is 0x000013 in Hello (V6-Bit, E-Bit, R-Bit)
Options is 0x000013 in DBD (V6-Bit, E-Bit, R-Bit)
Dead timer due in 00:00:34
Neighbor is up for 00:19:23
Index 1/1/2, retransmission queue length 0, number of retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec

R10#show ipv6 route

IPv6 Routing Table - default - 11 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
B - BGP, R - RIP, H - NHRP, I1 - ISIS L1
I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP
EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination
NDR - Redirect, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1
OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
a - Application

OI 2001:DB8:ACAD:1::/64 [110/6]
via FE80::B6A8:B9FF:FE01:AE51, GigabitEthernet0/0/0
OI 2001:DB8:ACAD:2::/64 [110/5]
via FE80::B6A8:B9FF:FE01:AE51, GigabitEthernet0/0/0
O 2001:DB8:ACAD:3::/64 [110/4]
via FE80::B6A8:B9FF:FE01:AE51, GigabitEthernet0/0/0
O 2001:DB8:ACAD:4::/64 [110/3]
via FE80::B6A8:B9FF:FE01:AE51, GigabitEthernet0/0/0
O 2001:DB8:ACAD:5::/64 [110/2]
via FE80::B6A8:B9FF:FE01:AE51, GigabitEthernet0/0/0
C 2001:DB8:ACAD:6::/64 [0/0]
via GigabitEthernet0/0/0, directly connected
L 2001:DB8:ACAD:6::2/128 [0/0]
via GigabitEthernet0/0/0, receive
C 2001:DB8:ACAD:7::/64 [0/0]

R10#show ip ospf database

OSPF Router with ID (10.10.10.10) (Process ID 1)

Router Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
2.2.2.2	2.2.2.2	215	0x8000003B	0x00F616	1
7.7.7.7	7.7.7.7	1954	0x80000006	0x00E447	2
8.8.8.8	8.8.8.8	51	0x80000005	0x00EA36	2
9.9.9.9	9.9.9.9	1720	0x80000005	0x00CC49	2
10.10.10.10	10.10.10.10	1687	0x80000003	0x0055A8	1

Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.3.1	2.2.2.2	215	0x80000002	0x00168C
192.168.4.2	8.8.8.8	51	0x80000002	0x001957
192.168.5.1	8.8.8.8	51	0x80000002	0x007CEB
192.168.6.1	9.9.9.9	1720	0x80000001	0x00A9B2

Summary Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.1.0	2.2.2.2	457	0x80000002	0x00A225
192.168.2.0	2.2.2.2	457	0x80000002	0x007B4F
192.168.7.0	10.10.10.10	1677	0x80000001	0x005551
192.168.8.0	10.10.10.10	1642	0x80000001	0x00663B

Router Link States (Area 1)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
10.10.10.10	10.10.10.10	1647	0x80000003	0x006794	1
11.11.11.11	11.11.11.11	1647	0x80000005	0x00D39B	2

Net Link States (Area 1)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.7.2	11.11.11.11	1647	0x80000001	0x009CAD

Summary Net Link States (Area 1)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.1.0	10.10.10.10	1687	0x80000001	0x00DBC8
192.168.2.0	10.10.10.10	1687	0x80000001	0x00B4F2

R10#show ip ospf int brief

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Gi0/0/0	1	0	192.168.6.2/30	1	BDR	1/1	
Gi0/0/1	1	1	192.168.7.1/30	1	BDR	1/1	

SWITCH 2:

S2(config-if)#do show ip ospf database

OSPF Router with ID (2.2.2.2) (Process ID 1)

Router Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
2.2.2.2	2.2.2.2	175	0x8000003B	0x00F616 1	
7.7.7.7	7.7.7.7	1917	0x80000006	0x00E447 2	
8.8.8.8	8.8.8.8	15	0x80000005	0x00EA36 2	
9.9.9.9	9.9.9.9	1686	0x80000005	0x00CC49 2	
10.10.10.10	10.10.10.10	1655	0x80000003	0x0055A8 1	

Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.3.1	2.2.2.2	175	0x80000002	0x00168C
192.168.4.2	8.8.8.8	15	0x80000002	0x001957
192.168.5.1	8.8.8.8	15	0x80000002	0x007CEB
192.168.6.1	9.9.9.9	1686	0x80000001	0x00A9B2

Summary Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.1.0	2.2.2.2	417	0x80000002	0x00A225
192.168.2.0	2.2.2.2	417	0x80000002	0x007B4F
192.168.7.0	10.10.10.10	1645	0x80000001	0x005551
192.168.8.0	10.10.10.10	1610	0x80000001	0x00663B

Router Link States (Area 51)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
1.1.1.1	1.1.1.1	458	0x80000049	0x002C62 2	
2.2.2.2	2.2.2.2	417	0x80000034	0x00FC18 1	

Net Link States (Area 51)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.2.1	1.1.1.1	708	0x8000002B	0x00CFC6

Summary Net Link States (Area 51)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.3.0	2.2.2.2	417	0x80000002	0x007059
192.168.4.0	2.2.2.2	1961	0x80000001	0x007157
192.168.5.0	2.2.2.2	1916	0x80000001	0x007056
192.168.6.0	2.2.2.2	1749	0x80000001	0x006F55
192.168.7.0	2.2.2.2	1641	0x80000001	0x006E54
192.168.8.0	2.2.2.2	1606	0x80000001	0x007F3E

S2(config-if)#do show ip ospf neigh detail

Neighbor 7.7.7.7, interface address 192.168.3.2

In the area 0 via interface GigabitEthernet1/0/2

Neighbor priority is 1, State is FULL, 6 state changes

DR is 192.168.3.1 BDR is 192.168.3.2

Options is 0x12 in Hello (E-bit, L-bit)

Options is 0x52 in DBD (E-bit, L-bit, O-bit)

LLS Options is 0x1 (LR)

Dead timer due in 00:00:36

Neighbor is up for 00:21:32

Index 1/1/2, retransmission queue length 0, number of retransmission 0

First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)

Last retransmission scan length is 0, maximum is 0

Last retransmission scan time is 0 msec, maximum is 0 msec

Neighbor 1.1.1.1, interface address 192.168.2.1, interface-id 39

In the area 51 via interface GigabitEthernet1/0/1

Neighbor priority is 1, State is FULL, 6 state changes

DR is 192.168.2.1 BDR is 192.168.2.2

Options is 0x12 in Hello (E-bit, L-bit)

Options is 0x52 in DBD (E-bit, L-bit, O-bit)

LLS Options is 0x1 (LR)

Dead timer due in 00:00:36

Neighbor is up for 23:24:50

Index 1/1/1, retransmission queue length 0, number of retransmission 0

First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)

Last retransmission scan length is 0, maximum is 0

Last retransmission scan time is 0 msec, maximum is 0 msec

S2(config-if)#do show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP

n - NAT, Ni - NAT inside, No - NAT9/30/2025 outside, Nd - NAT DIA

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

H - NHRP, G - NHRP registered, g - NHRP registration summary

o - ODR, P - periodic downloaded static route, l - LISP

a - application route

+ - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

O 192.168.1.0/24 [110/2] via 192.168.2.1, 00:35:06, GigabitEthernet1/0/1

192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.2.0/30 is directly connected, GigabitEthernet1/0/1

L 192.168.2.2/32 is directly connected, GigabitEthernet1/0/1

192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.3.0/30 is directly connected, GigabitEthernet1/0/2

L 192.168.3.1/32 is directly connected, GigabitEthernet1/0/2

192.168.4.0/30 is subnetted, 1 subnets

O 192.168.4.0 [110/2] via 192.168.3.2, 00:30:02, GigabitEthernet1/0/2

192.168.5.0/30 is subnetted, 1 subnets

O 192.168.5.0 [110/3] via 192.168.3.2, 00:30:02, GigabitEthernet1/0/2

192.168.6.0/30 is subnetted, 1 subnets

O 192.168.6.0 [110/4] via 192.168.3.2, 00:27:15, GigabitEthernet1/0/2

192.168.7.0/30 is subnetted, 1 subnets

O IA 192.168.7.0 [110/5] via 192.168.3.2, 00:25:27, GigabitEthernet1/0/2

O IA 192.168.8.0/24 [110/6] via 192.168.3.2, 00:24:52, GigabitEthernet1/0/2

S2(config-if)#do show ipv6 ospf neigh detail

OSPFv3 Router with ID (2.2.2.2) (Process ID 1)

Neighbor 7.7.7.7

In the area 0 via interface GigabitEthernet1/0/2

Neighbor: interface-id 6, link-local address FE80::B6A8:B9FF:FE01:B510

Neighbor priority is 1, State is FULL, 6 state changes

DR is 7.7.7.7 BDR is 2.2.2.2

Options is 0x000013 in Hello (V6-Bit, E-Bit, R-Bit)

Options is 0x000013 in DBD (V6-Bit, E-Bit, R-Bit)

Dead timer due in 00:00:35

Neighbor is up for 00:14:38

Index 1/1/2, retransmission queue length 0, number of retransmission 0

First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)

Last retransmission scan length is 0, maximum is 0

Last retransmission scan time is 0 msec, maximum is 0 msec

Neighbor 1.1.1.1

In the area 51 via interface GigabitEthernet1/0/1

Neighbor: interface-id 39, link-local address FE80::BED2:95FF:FE3C:4AE4

Neighbor priority is 1, State is FULL, 6 state changes

DR is 1.1.1.1 BDR is 2.2.2.2

Options is 0x000013 in Hello (V6-Bit, E-Bit, R-Bit)

Options is 0x000013 in DBD (V6-Bit, E-Bit, R-Bit)

Dead timer due in 00:00:36

Neighbor is up for 00:15:05

Index 1/1/1, retransmission queue length 0, number of retransmission 0

First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)

Last retransmission scan length is 0, maximum is 0

Last retransmission scan time is 0 msec, maximum is 0 msec

S2#show ipv6 route

IPv6 Routing Table - default - 11 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

R - RIP, D - EIGRP, EX - EIGRP external, ND - ND Default

NDp - ND Prefix, DCE - Destination, NDr - Redirect, O - OSPF Intra

OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1

ON2 - OSPF NSSA ext 2

O 2001:DB8:ACAD:1::/64 [110/2]

via FE80::BED2:95FF:FE3C:4AE4, GigabitEthernet1/0/1

C 2001:DB8:ACAD:2::/64 [0/0]

via GigabitEthernet1/0/1, directly connected

L 2001:DB8:ACAD:2::1/28 [0/0]

via GigabitEthernet1/0/1, receive

C 2001:DB8:ACAD:3::/64 [0/0]

via GigabitEthernet1/0/2, directly connected

L 2001:DB8:ACAD:3::1/28 [0/0]

via GigabitEthernet1/0/2, receive

O 2001:DB8:ACAD:4::/64 [110/2]

via FE80::B6A8:B9FF:FE01:B510, GigabitEthernet1/0/2

O 2001:DB8:ACAD:5::/64 [110/3]

via FE80::B6A8:B9FF:FE01:B510, GigabitEthernet1/0/2

O 2001:DB8:ACAD:6::/64 [110/4]

via FE80::B6A8:B9FF:FE01:B510, GigabitEthernet1/0/2

OI 2001:DB8:ACAD:7::/64 [110/5]

via FE80::B6A8:B9FF:FE01:B510, GigabitEthernet1/0/2

OI 2001:DB8:ACAD:8::/64 [110/6]

via FE80::B6A8:B9FF:FE01:B510, GigabitEthernet1/0/2

L FF00::/8 [0/0]

via Null0, receive

S2#show ip ospf int brief

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Gi1/0/2	1	0	192.168.3.1/30	1	DR	1/1	
Gi1/0/1	1	51	192.168.2.2/30	1	BDR	1/1	

SWITCH 1:**S1#show ip route**

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP

n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

H - NHRP, G - NHRP registered, g - NHRP registration summary

o - ODR, P - periodic downloaded static route, l - LISP

a - application route

+ - replicated route, % - next hop override, p - overrides from PfR

& - replicated local route overrides by connected

Gateway of last resort is not set

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.1.0/24 is directly connected, GigabitEthernet1/0/2

L 192.168.1.1/32 is directly connected, GigabitEthernet1/0/2

192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.2.0/30 is directly connected, GigabitEthernet1/0/1

L 192.168.2.1/32 is directly connected, GigabitEthernet1/0/1

192.168.3.0/30 is subnetted, 1 subnets

O IA 192.168.3.0 [110/2] via 192.168.2.2, 00:53:43, GigabitEthernet1/0/1

192.168.4.0/30 is subnetted, 1 subnets

O IA 192.168.4.0 [110/3] via 192.168.2.2, 00:49:23, GigabitEthernet1/0/1

192.168.5.0/30 is subnetted, 1 subnets

O IA 192.168.5.0 [110/4] via 192.168.2.2, 00:48:39, GigabitEthernet1/0/1

192.168.6.0/30 is subnetted, 1 subnets

O IA 192.168.6.0 [110/5] via 192.168.2.2, 00:45:52, GigabitEthernet1/0/1

192.168.7.0/30 is subnetted, 1 subnets

O IA 192.168.7.0 [110/6] via 192.168.2.2, 00:44:04, GigabitEthernet1/0/1

O IA 192.168.8.0/24 [110/7] via 192.168.2.2, 00:43:29, GigabitEthernet1/0/1

S1#show ipv6 route

IPv6 Routing Table - default - 11 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

R - RIP, D - EIGRP, EX - EIGRP external, ND - ND Default

NDp - ND Prefix, DCE - Destination, NDr - Redirect, O - OSPF Intra

OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1

ON2 - OSPF NSSA ext 2

C 2001:DB8:ACAD:1::/64 [0/0]

via GigabitEthernet1/0/2, directly connected

L 2001:DB8:ACAD:1::1/128 [0/0]

via GigabitEthernet1/0/2, receive

C 2001:DB8:ACAD:2::/64 [0/0]

via GigabitEthernet1/0/1, directly connected

L 2001:DB8:ACAD:2::1/128 [0/0]

via GigabitEthernet1/0/1, receive

OI 2001:DB8:ACAD:3::/64 [110/2]

via FE80::B28B:D0FF:FED6:C64, GigabitEthernet1/0/1

OI 2001:DB8:ACAD:4::/64 [110/3]

via FE80::B28B:D0FF:FED6:C64, GigabitEthernet1/0/1

OI 2001:DB8:ACAD:5::/64 [110/4]

via FE80::B28B:D0FF:FED6:C64, GigabitEthernet1/0/1

OI 2001:DB8:ACAD:6::/64 [110/5]

via FE80::B28B:D0FF:FED6:C64, GigabitEthernet1/0/1

OI 2001:DB8:ACAD:7::/64 [110/6]

via FE80::B28B:D0FF:FED6:C64, GigabitEthernet1/0/1

OI 2001:DB8:ACAD:8::/64 [110/7]

via FE80::B28B:D0FF:FED6:C64, GigabitEthernet1/0/1

L FF00::/8 [0/0]

via Null0, receive

S1#show ip ospf neigh detail

Neighbor 2.2.2.2, interface address 192.168.2.2, interface-id 39
In the area 51 via interface GigabitEthernet1/0/1
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.2.1 BDR is 192.168.2.2
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:35
Neighbor is up for 23:57:39
Index 1/1/1, retransmission queue length 0, number of retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec

S1#show ip ospf int brief

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Gi1/0/1	1	51	192.168.2.1/30	1	DR	1/1	
Gi1/0/2	1	51	192.168.1.1/24	1	DR	0/0	

S1#show ipv6 neigh detail

IPv6 Address	TRLV	Age	Link-layer	Addr	State	Interface
2001:DB8:ACAD:1::2	0	28	04d9.c8ba.2574	STALE	Gi1/0/2	
2001:DB8:ACAD:1:705D:3E65:F064:1A11	0	0	04d9.c8ba.2574	REACH	Gi1/0/2	
FE80::19A5:5D31:E6D0:6DA	0	0	04d9.c8ba.2574	REACH	Gi1/0/2	
FE80::B28B:D0FF:FED6:C64	0	36	b08b.d0d6.0c64	STALE	Gi1/0/1	

ROUTER 11:

R11#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
Gateway of last resort is not set
O IA 192.168.1.0/24 [110/7] via 192.168.7.1, 00:48:39, GigabitEthernet0/0/0
192.168.2.0/30 is subnetted, 1 subnets
O IA 192.168.2.0 [110/6] via 192.168.7.1, 00:48:39, GigabitEthernet0/0/0
192.168.3.0/30 is subnetted, 1 subnets
O IA 192.168.3.0 [110/5] via 192.168.7.1, 00:48:39, GigabitEthernet0/0/0
192.168.4.0/30 is subnetted, 1 subnets
O IA 192.168.4.0 [110/4] via 192.168.7.1, 00:48:39, GigabitEthernet0/0/0
192.168.5.0/30 is subnetted, 1 subnets
O IA 192.168.5.0 [110/3] via 192.168.7.1, 00:48:39, GigabitEthernet0/0/0
192.168.6.0/30 is subnetted, 1 subnets
O IA 192.168.6.0 [110/2] via 192.168.7.1, 00:48:39, GigabitEthernet0/0/0
192.168.7.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.7.0/30 is directly connected, GigabitEthernet0/0/0
L 192.168.7.2/32 is directly connected, GigabitEthernet0/0/0
192.168.8.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.8.0/24 is directly connected, GigabitEthernet0/0/1
L 192.168.8.1/32 is directly connected, GigabitEthernet0/0/1

R11#show ip ospf int brief

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Gi0/0/0	1	1	192.168.7.2/30	1	DR	1/1	
Gi0/0/1	1	1	192.168.8.1/24	1	DR	0/0	

R11# show ipv6 route

IPv6 Routing Table - default - 11 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external
ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application
OI 2001:DB8:ACAD:1::/64 [110/7]
via FE80::521C:B0FF:FE2D:7101, GigabitEthernet0/0/0
OI 2001:DB8:ACAD:2::/64 [110/6]
via FE80::521C:B0FF:FE2D:7101, GigabitEthernet0/0/0
OI 2001:DB8:ACAD:3::/64 [110/5]
via FE80::521C:B0FF:FE2D:7101, GigabitEthernet0/0/0
OI 2001:DB8:ACAD:4::/64 [110/4]
via FE80::521C:B0FF:FE2D:7101, GigabitEthernet0/0/0
OI 2001:DB8:ACAD:5::/64 [110/3]
via FE80::521C:B0FF:FE2D:7101, GigabitEthernet0/0/0
OI 2001:DB8:ACAD:6::/64 [110/2]
via FE80::521C:B0FF:FE2D:7101, GigabitEthernet0/0/0
C 2001:DB8:ACAD:7::/64 [0/0]
via GigabitEthernet0/0/0, directly connected
L 2001:DB8:ACAD:7::2/128 [0/0]
via GigabitEthernet0/0/0, receive
C 2001:DB8:ACAD:8::/64 [0/0]
via GigabitEthernet0/0/1, directly connected
L 2001:DB8:ACAD:8::1/128 [0/0]
via GigabitEthernet0/0/1, receive
L FF00::/8 [0/0]
via Null0, receive

R11#show ip ospf neigh detail

```
Neighbor 10.10.10.10, interface address 192.168.7.1
In the area 1 via interface GigabitEthernet0/0/0
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.7.2 BDR is 192.168.7.1
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:34
Neighbor is up for 00:50:31
Index 1/1/1, retransmission queue length 0, number of retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec
```

R11#show ipv6 ospf neigh de

```
OSPFv3 Router with ID (11.11.11.11) (Process ID 1)
Neighbor 10.10.10.10
In the area 1 via interface GigabitEthernet0/0/0
Neighbor: interface-id 7, link-local address FE80::521C:B0FF:FE2D:7101
Neighbor priority is 1, State is FULL, 6 state changes
DR is 10.10.10.10 BDR is 11.11.11.11
Options is 0x000013 in Hello (V6-Bit, E-Bit, R-Bit)
Options is 0x000013 in DBD (V6-Bit, E-Bit, R-Bit)
Dead timer due in 00:00:38
Neighbor is up for 00:43:46
Index 1/1/1, retransmission queue length 0, number of retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec
```

SWITCH 1 SHOW RUN:

```
S1#show run
Current configuration : 9345 bytes
!
! Last configuration change at 14:56:49 UTC Wed Sep 24 2025
!
version 17.3
service timestamps debug datetime msec
service timestamps log datetime msec
! Call-home is enabled by Smart-Licensing.
service call-home
platform punt-keepalive disable-kernel-core
!
hostname S1
!
!
vrf definition Mgmt-vrf
!
address-family ipv4
exit-address-family
!
address-family ipv6
exit-address-family
!
no aaa new-model
switch 1 provision c9200l-24t-4g
!
ip routing
!
login on-success log
ipv6 unicast-routing
no device-tracking logging theft
!
crypto pki trustpoint TP-self-signed-2577419285
enrollment selfsigned
subject-name cn=IOS-Self-Signed-Certificate-2577419285
revocation-check none
rsa-keypair TP-self-signed-2577419285
!
crypto pki trustpoint SLA-TrustPoint
enrollment pkcs12
revocation-check crl
!
crypto pki certificate chain TP-self-signed-2577419285
certificate self-signed 01
30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101 05050030
31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D 43657274
69666963 6174652D 32353737 34313932 3835301E 170D3235 30393234 31343433
31335A17 0D333530 39323431 34343331 335A3031 312F302D 06035504 03132649
4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D32 35373734
31393238 35308201 22300D06 092A8648 86F70D01 01010500 0382010F 00308201
0A028201 0100BF8F C6D325BF 986B201D 487C7B73 0F16CFB0 8369A1C0 34E91FCF
3C18A99A 3D755EC4 6C2E66B5 5C3A00D4 FC980261 15E7D936 07481731 FD76A7B5
```


744D8EA9 D561AC0B D6F115E7 61CD2A17 D91E61D2 FF8C00FD 73F213BD 2974F619
B175D974 1FCCB579 ECA51034 B16B55F2 CC91BACA 30B6DC9D 0E25A951 95F18224
AF6BA461 EE02A0BB 81DCBC42 E4BCECF5 5D7FD12C DB2D7383 8506483E A4362F2B
73D526FA 25D89C93 A851DD93 532FDDA9 42E6602E C03044E7 8A21D675 ADF95A18
6C7C18C9 F815420D A481D4DA 2D23FB7D 6BB46BAF 6D740A19 28216F58 EE8D9136
F26D58BD 66C63282 A181DA32 24BD2188 679B6BA5 3BE01A5D 6DE9D713 8BB04339
CF573E43 971B0203 010001A3 53305130 0F060355 1D130101 FF040530 030101FF
301F0603 551D2304 18301680 14DCF50A B4F70B82 D2B1FFDD E610E46A 960B379A
7C301D06 03551D0E 04160414 DCF50AB4 F70B82D2 B1FFDDE6 10E46A96 0B379A7C
300D0609 2A864886 F70D0101 05050003 82010100 A107032D 5A0C16E4 83F962ED
C66C600F CD7D43E7 EA14DB50 3EE9B8B9 7B2EA40A 005AE896 BDD75BA0 63953D05
3D63F65D 4E5AD350 D79DC07B 3B424527 91A6A28C CD2B519C 8DCCA9C4 36699103
E80A8E4F 2FAE71D2 25F41F03 70DD7069 51F7E47F A75C743D 8EB96B73 F4AAB96C
5ACD96F4 0444F453 283FD57F 99863177 22B8F884 A4C4B588 5E07543B 92FF92FE
9FBA131A BE21EF6F 27594921 FB98D6A5 44073ECA 2D002E9F 7F10C9A0 E0120DC2
CDC7980B 55E73C0B 41112F28 B7F52B24 8EDFF1CF E11BF2DA 23D4055A 1045C740
8DF787EC F015A4AF 2B54CBA9 B0570C1C 26D351D7 37EB135F E204DD27 2D13C165
94B6A16A E4C99AC3 377D02E3 CDEA2D43 AD84FCBC

quit

crypto pki certificate chain SLA-TrustPoint

certificate ca 01

30820321 30820209 A0030201 02020101 300D0609 2A864886 F70D0101 0B050030
32310E30 0C060355 040A1305 43697363 6F312030 1E060355 04031317 43697363
6F204C69 63656E73 696E6720 526F6F74 20434130 1E170D31 33303533 30313934
3834375A 170D3338 30353330 31393438 34375A30 32310E30 0C060355 040A1305
43697363 6F312030 1E060355 04031317 43697363 6F204C69 63656E73 696E6720
526F6F74 20434130 82012230 0D06092A 864886F7 0D010101 05000382 010F0030
82010A02 82010100 A6BCBD96 131E05F7 145EA72C 2CD686E6 17222EA1 F1EFF64D
CBB4C798 212AA147 C655D8D7 9471380D 8711441E 1AAF071A 9CAE6388 8A38E520
1C394D78 462EF239 C659F715 B98C0A59 5BBB5CBD 0CFEBAE3 700A8BF7 D8F256EE
4AA4E80D DB6FD1C9 60B1FD18 FFC69C96 6FA68957 A2617DE7 104FDC5F EA2956AC
7390A3EB 2B5436AD C847A2C5 DAB553EB 69A9A535 58E9F3E3 C0BD23CF 58BD7188
68E69491 20F320E7 948E71D7 AE3BCC84 F10684C7 4BC8E00F 539BA42B 42C68BB7
C7479096 B4CB2D62 EA2F505D C7B062A4 6811D95B E8250FC4 5D5D5FB8 8F27D191
C55F0D76 61F9A4CD 3D992327 A8BB03BD 4E6D7069 7CBADF8B DF5F4368 95135E44
DFC7C6CF 04DD7FD1 02030100 01A34230 40300E06 03551D0F 0101FF04 04030201
06300F06 03551D13 0101FF04 05300301 01FF301D 0603551D 0E041604 1449DC85
4B3D31E5 1B3E6A17 606AF333 3D3B4C73 E8300D06 092A8648 86F70D01 010B0500
03820101 00507F24 D3932A66 86025D9F E838AE5C 6D4DF6B0 49631C78 240DA905
604EDCDE FF4FED2B 77FC460E CD636FDB DD44681E 3A5673AB 9093D3B1 6C9E3D8B
D98987BF E40CBD9E 1AECA0C2 2189BB5C 8FA85686 CD98B646 5575B146 8DFC66A8
467A3DF4 4D565700 6ADF0F0D CF835015 3C04FF7C 21E878AC 11BA9CD2 55A9232C
7CA7B7E6 C1AF74F6 152E99B7 B1FCF9BB E973DE7F 5BDDEB86 C71E3B49 1765308B
5FB0DA06 B92AFE7F 494E8A9E 07B85737 F3A58BE1 1A48A229 C37C1E69 39F08678
80DDCD16 D6BACECA EEBC7CF9 8428787B 35202CDC 60E4616A B623CDBD 230E3AFB
418616A9 4093E049 4D10AB75 27E86F73 932E35B5 8862FDAE 0275156F 719BB2F0
D697DF7F 28

!

license boot level network-essentials addon dna-essentials

!

diagnostic bootup level minimal

!

spanning-tree mode rapid-pvst

spanning-tree extend system-id

memory free low-watermark processor 10308

!

redundancy

mode sso

!

transceiver type all

monitoring

!

class-map match-any system-cpp-police-ewlc-control

description EWLC Control

class-map match-any system-cpp-police-topology-control

description Topology control

class-map match-any system-cpp-police-sw-forward

description Sw forwarding, L2 LVX data packets, LOGGING, Transit Traffic

class-map match-any system-cpp-default

description EWLC data, Inter FED Traffic

class-map match-any system-cpp-police-sys-data

description Openflow, Exception, EGR Exception, NFL Sampled Data, RPF Failed

class-map match-any system-cpp-police-punt-webauth

description Punt Webauth

class-map match-any system-cpp-police-l2lvx-control

```

    description L2 LVX control packets
class-map match-any system-cpp-police-forus
    description Forus Address resolution and Forus traffic
class-map match-any system-cpp-police-multicast-end-station
    description MCAST END STATION
class-map match-any system-cpp-police-high-rate-app
    description High Rate Applications
class-map match-any system-cpp-police-multicast
    description MCAST Data
class-map match-any system-cpp-police-l2-control
    description L2 control
class-map match-any system-cpp-police-dot1x-auth
    description DOT1X Auth
class-map match-any system-cpp-police-data
    description ICMP redirect, ICMP_GEN and BROADCAST
class-map match-any system-cpp-police-stackwise-virt-control
    description Stackwise Virtual OOB
class-map match-any non-client-nrt-class
class-map match-any system-cpp-police-routing-control
    description Routing control and Low Latency
class-map match-any system-cpp-police-protocol-snooping
    description Protocol snooping
class-map match-any system-cpp-police-dhcp-snooping
    description DHCP snooping
class-map match-any system-cpp-police-ios-routing
    description L2 control, Topology control, Routing control, Low Latency
class-map match-any system-cpp-police-system-critical
    description System Critical and Gold Pkt
class-map match-any system-cpp-police-ios-feature
    description
ICMPGEN,BROADCAST,ICMP,L2LVXCntrl,ProtoSnoop,PuntWebauth,MCASTData,Transit,DOT1XAuth,Swfwd,LOGGING,L2LVXData,ForusTraffic,ForusARP,Mc
astEndStn,Openflow,Exception,EGRExcption,NflSampled,RpfFailed
!
policy-map system-cpp-policy
!
interface GigabitEthernet0/0
vrf forwarding Mgmt-vrf
no ip address
shutdown
negotiation auto
!
interface GigabitEthernet1/0/1
no switchport
ip address 192.168.2.1 255.255.255.252
ip ospf 1 area 1
ipv6 address 2001:DB8:ACAD:2::1/64
ipv6 ospf 1 area 1
!
interface GigabitEthernet1/0/2
no switchport
ip address 192.168.1.1 255.255.255.252
ip ospf 1 area 1
ipv6 address 2001:DB8:ACAD:1::1/64
ipv6 ospf 1 area 1
!
interface GigabitEthernet1/0/3
!
interface GigabitEthernet1/0/4
!
interface GigabitEthernet1/0/5
!
interface GigabitEthernet1/0/6
!
interface GigabitEthernet1/0/7
!
interface GigabitEthernet1/0/8
!
interface GigabitEthernet1/0/9
!
interface GigabitEthernet1/0/10
!
interface GigabitEthernet1/0/11
!
interface GigabitEthernet1/0/12
!

```

```
interface GigabitEthernet1/0/13
!
interface GigabitEthernet1/0/14
!
interface GigabitEthernet1/0/15
!
interface GigabitEthernet1/0/16
!
interface GigabitEthernet1/0/17
!
interface GigabitEthernet1/0/18
!
interface GigabitEthernet1/0/19
!
interface GigabitEthernet1/0/20
!
interface GigabitEthernet1/0/21
!
interface GigabitEthernet1/0/22
!
interface GigabitEthernet1/0/23
!
interface GigabitEthernet1/0/24
!
interface GigabitEthernet1/1/1
!
interface GigabitEthernet1/1/2
!
interface GigabitEthernet1/1/3
!
interface GigabitEthernet1/1/4
!
interface Vlan1
no ip address
shutdown
!
router ospf 1
router-id 1.1.1.1
!
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
!
!
ipv6 router ospf 1
router-id 1.1.1.1
!
!
!
control-plane
service-policy input system-cpp-policy
!
!
line con 0
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
transport input ssh
line vty 5 15
login
transport input ssh
!
call-home
! If contact email address in call-home is configured as sch-smart-licensing@cisco.com
! the email address configured in Cisco Smart License Portal will be used as contact email address to send SCH notifications.
contact-email-addr sch-smart-licensing@cisco.com
profile "CiscoTAC-1"
active
destination transport-method http
!
end
```

ROUTER 7 SHOW RUN:

```
R7#show run
Building configuration...
Current configuration : 1628 bytes
!
! Last configuration change at 19:00:21 UTC Wed Sep 24 2025
!
version 15.5
service timestamps debug datetime msec
service timestamps log datetime msec
no platform punt-keepalive disable-kernel-core
!
hostname R7
!
boot-start-marker
boot-end-marker
!
!
vrf definition Mgmt-intf
!
address-family ipv4
exit-address-family
!
address-family ipv6
exit-address-family
!
!
no aaa new-model
!
ipv6 unicast-routing
!
subscriber templating
vtp domain cisco
vtp mode transparent
multilink bundle-name authenticated
!
license udi pid ISR4321/K9 sn FDO21491FHX
!
spanning-tree extend system-id
!
redundancy
mode none
!
vlan internal allocation policy ascending
!
interface GigabitEthernet0/0/0
ip address 192.168.3.2 255.255.255.252
ip ospf 1 area 0
negotiation auto
ipv6 address 2001:DB8:ACAD:3::2/64
ipv6 ospf 1 area 0
!
interface GigabitEthernet0/0/1
ip address 192.168.4.1 255.255.255.252
ip ospf 1 area 0
negotiation auto
ipv6 address 2001:DB8:ACAD:4::1/64
ipv6 ospf 1 area 0
!
interface GigabitEthernet0/1/0
no ip address
shutdown
negotiation auto
!
interface GigabitEthernet0/1/1
no ip address
shutdown
negotiation auto
!
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
```



```

!
interface Vlan1
no ip address
shutdown
!
router ospf 1
router-id 7.7.7.7
!
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
!
!
ipv6 router ospf 1
router-id 7.7.7.7
!
control-plane
!
line con 0
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
!
!
end

```

Problems

The primary issue that we encountered during the lab was related to wiring. We spent a lot of time troubleshooting what we initially thought was a configuration error with OSPF. However, the issue was ultimately traced to a layer 1 problem: router labels were misread, resulting in cables being connected to incorrect ports. Once the wiring was corrected and aligned with the proper router assignments, OSPF operated as intended.

Conclusion

This lab demonstrated the configuration and operation of multi-area OSPF using both IPv4 and IPv6 across a network of routers and Layer 3 switches. By dividing the topology into multiple areas, we observed how OSPF reduces routing table size and improves scalability. Despite initial difficulties caused by wiring errors at the physical layer, the successful correction of these issues allowed OSPF to function as intended, with proper route exchange and end-to-end connectivity. Overall, the lab reinforced key concepts of link-state routing, area design, and the importance of accurate physical setup in ensuring reliable protocol operation.

Teacher Signoff Page of Lab Completed

